

Using Artificial Neural Networks and Kolmogorov-Zurbenko Filtering to Predict Future Sunspot Numbers and Temperatures



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ABSTRACT

Predicting sunspot numbers aids with the prediction of space weather, induced ionospheric turbulence, satellite lifespans, communication disruptions, coronal mass ejections, astronaut health risks, and geomagnetic storm activity. Predicting future temperatures will aid with climate studies and weather predictions. This study addresses the need for an efficient, quick-to-train, and low-cost deep learning model to perform sunspot number and temperature predictions with competitive accuracy. Univariate and multivariate models were trained on daily and monthly observations of both data sets. KZ filtering was employed to smooth out noise in both data sets. Models trained on KZ-filtered data showed dramatic increases in training efficiency while maintaining high accuracies. The multivariate ANNs trained on KZ-filtered sunspot numbers and temperatures showed a **70.082% decrease** and a **69.041% decrease** in steps required to converge during training, respectively. The sunspot number ANN and temperature ANN had **MSEs of 0.00385 and 0.003854**, respectively. The sunspot number and temperature models produced reliable predictions for 100 days and 200 days in the future, respectively. Hidden patterns of both data sets were revealed using KZ filtering in combination with a wavelet analysis. Models were trained on hidden patterns and they achieved high accuracy. This study establishes that KZ filtering can be used with deep learning to significantly boost training efficiency by dramatically reducing the number of steps required for a model to converge during training. Wavelet analysis in combination with KZ filtering can be used to detect hidden patterns for any data set.

METHODOLOGY

- The sunspot data set (**75,484 observations**) was downloaded using the **data.table package** in R from the following URL, obtained from the Hofstra Geoinformatics Lab: http://www.sidc.be/silso/DATA/SN_d_tot_V2.0.csv (Figure 2a).
- The temperature data set (**27,809 observations**) was downloaded using the **GSDOR package** in R. Historical temperatures for Long Island were downloaded from stations 744860-94789 and 999999-94789 (figure 2a).
- Both data sets had missing dates, duplicate dates, and outliers removed. A linear interpolation with the imputeTS package was performed to fill in N/A values (Figure 2a).
- Both data sets had observations passed through a KZ(31, 3) filter to smooth the data to monthly data.
- Observations were rescaled to 0-1 for optimal performance (Figure 2b).
- The **neuralnet package** was used to train, test, and extract future predictions from 2 ANNs for both data sets (Figure 2b).
- Variables were lagged three times to improve multivariate deep learning performance.
- The train : test ratio was **80:20** (Figure 2b).
- A wavelet analysis was performed on both data sets. The period with the highest wavelet power level was passed into the KZ filter to reveal hidden patterns in the data set (Figure 2c).
- 2 more ANNs were created and trained on hidden patterns (Figure 2c).
- All models were evaluated graphically and using MSE as a numerical performance metric (figure 2c).

KZ-Filter equation utilized for the moving average adapted from Zurbenko (1986); package kza (Close et al. 2020).

$$Y_t = \frac{1}{m} \sum_{j=-k}^k (X_{t+j})$$

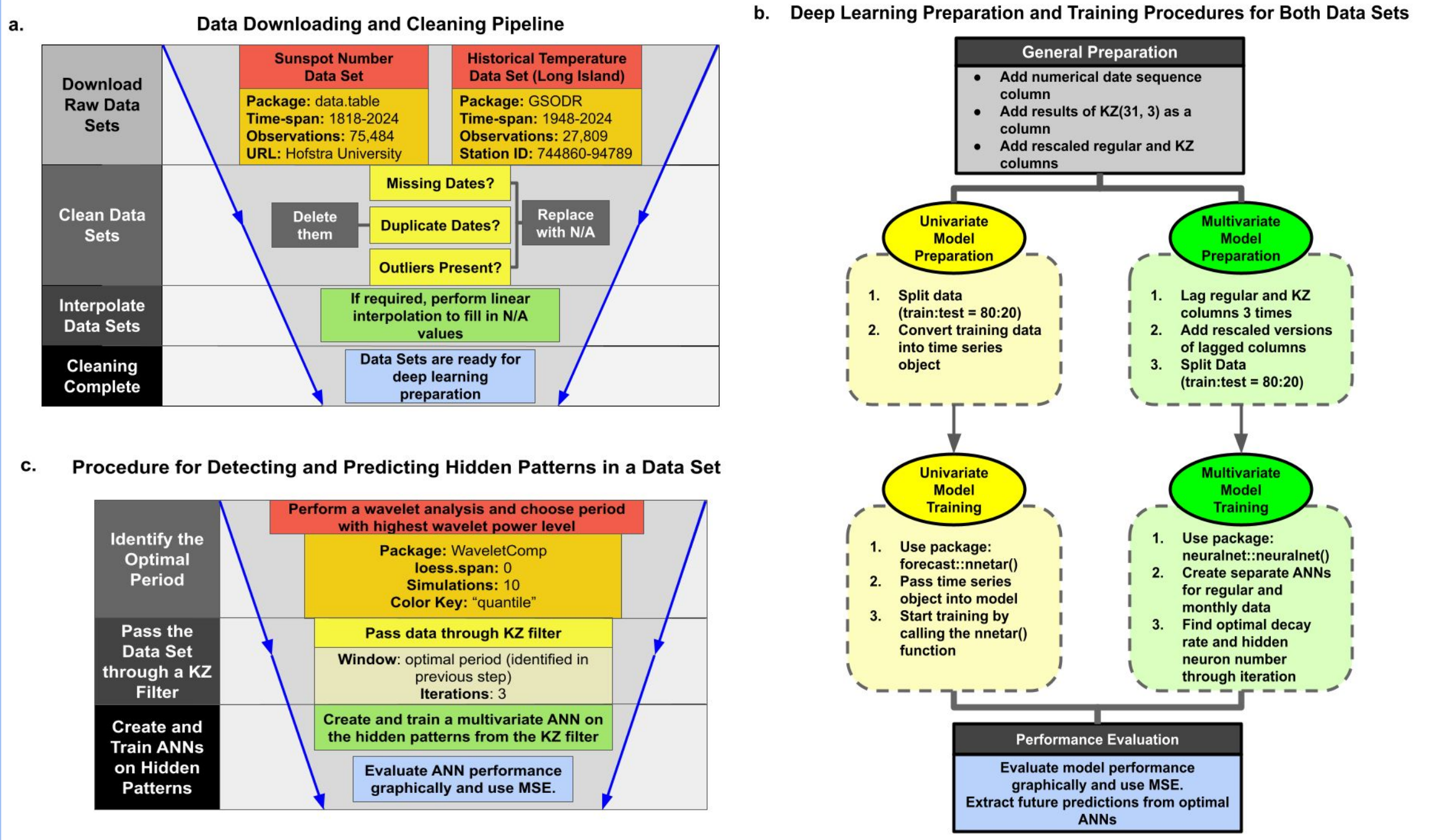


Figure 2. Methodology for the study. a. Data Downloading and Cleaning Pipeline. Both data sets were separately processed. b. Deep Learning Preparation and Training Procedures for Both Data Sets. The univariate ANN showed poor performance, so it was discarded. c. Procedure for Detecting and Predicting Hidden Patterns in a Data Set.

DISCUSSION

Sunspot Number Data Analysis

- The model trained on daily sunspot number data could not converge within a reasonable step max (100,000 steps) due to pattern complexity, so it was discarded.
- The model trained on KZ (31, 3) data converged in **26,918 steps** and showed a **70.082% decrease** in the steps required to converge during training (Figure 3B).
- The model achieved an **MSE of 0.00385**, which is competitive (Figure 3B).
- The model achieved high accuracy during training and produced reliable predictions for **100 days in the future**, aiding with space weather predictions (Figures 3C & 3D).
- The optimal period (4015 days) identified by the wavelet analysis makes sense when the 11 year solar cycle is considered (365 days* 11 years = 4015 days) (Figure 3E).
- The hidden patterns confirm speculations that we are passing through a **minimum in the Gleissberg Centennial Cycle**, showcased by the orange line (Figure 3F).
- The hidden pattern ANN converged in **11,221 steps** and showed a **58.341% decrease** in the number of steps required to converge during training (Figure 3G).
- The model achieved an **MSE of 0.002145** during training (Figures 3G & 3H).

Historical Temperature (Long Island) Data Analysis

- The model trained on daily temperature data could not converge within a reasonable stepmax (100,000 steps) due to pattern complexity, so it was discarded.
- The model trained on KZ (31, 3) data converged in **30,959 steps** and showed a **69.041% decrease** in the steps required to converge during training (Figure 4B).
- The model retained high accuracy, with **MSE of 0.003854** during training (Figure 4C).
- The model produced reliable temperature predictions for **200 days in the future**, which can aid with climate studies (Figure 4D).
- The optimal period (365 days) identified by the wavelet analysis makes sense when annual temperature fluctuations due to seasons are considered (Figure 4E).
- The hidden patterns showcase temperature rises in recent years (Figure 4F).
- The hidden pattern ANN converged in **7,973 steps** and showed a **74.247% decrease** in the number of steps required to converge during training (Figure 4G).
- The model achieved an **MSE of 0.008438** during training (Figures 4G & 4H).

CONCLUSIONS

- Prediction accuracy is retained** when a model is trained efficiently on smoothed data. The sunspot number and temperature predictions are useful in their fields.
- Multivariate ANNs can generate **reliable future predictions** for large data sets.
- KZ Filtering can be used in combination with deep learning to **significantly boost training efficiency** by **dramatically reducing the number of steps** required for a model to converge during training.
- Parameters of the KZ function can be adjusted to obtain **predictions of hidden patterns or efficient predictions of regular data**.
- A wavelet analysis in combination with KZ filtering can be used to **detect and predict future hidden patterns of any data set**.

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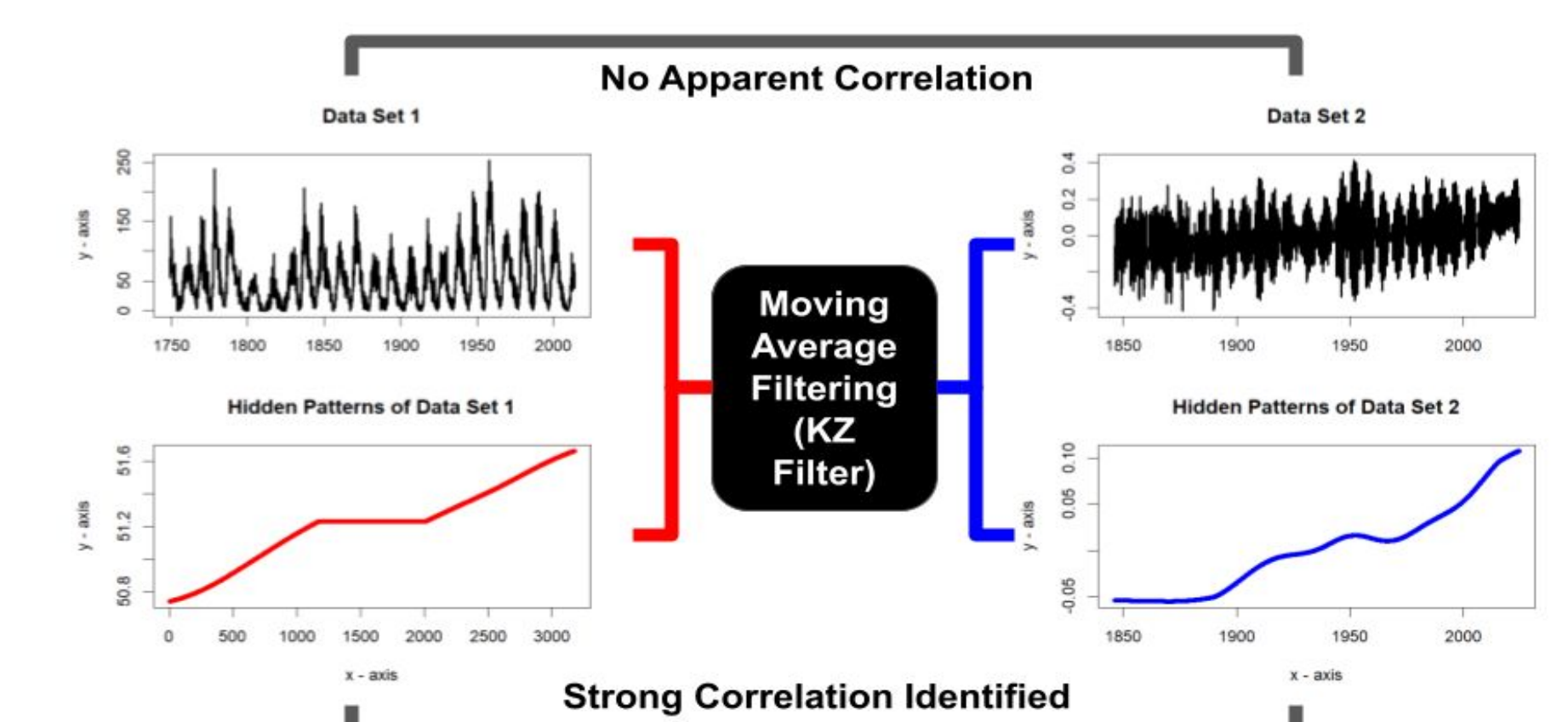
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INTRODUCTION

- Sunspot number predictions aid with the prediction of Coronal Mass Ejections, Solar Flares, and the Earth's geomagnetic storm activity (Rodriguez, 2024).
- Sunspot number predictions** aid with the prediction of space weather, satellite lifespans, communication disruptions, induced turbulence in the ionosphere, and astronaut health risks (Mourato, 2024).
- Kolmogorov-Zurbenko (KZ) Filtering** is a powerful moving average filter that smooths out noise and keeps the tail of a dataframe intact (Figure 1a & 1b).
- Artificial Neural Networks (ANNs)** are computational networks inspired by neurons in the human brain. An ANN is a supervised deep learning model that has a wide variety of applications in fields of prediction.
- Wavelet analysis** is a mathematical function that decomposes a signal into variable periods, taking into account changes in periodicities over time.
- Deep learning model training can be expensive and time consuming for large data sets. **A way to dramatically improve training efficiency is required.**

a. Sample of KZ Filtering to Identify Hidden Pattern Correlations



b. KZ Filtering vs. Other Moving Average Filters for Time Series Data

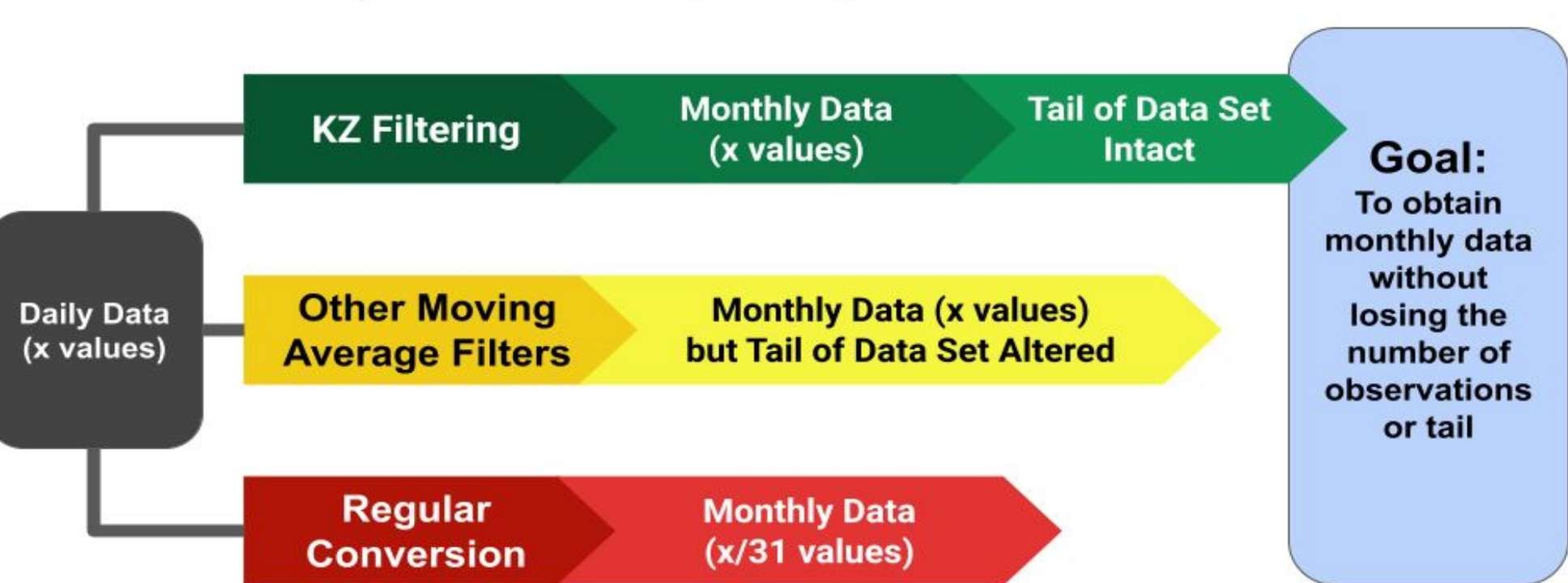


Figure 1. The applications of KZ filtering for identifying hidden patterns and finding correlations in time series data sets. a. KZ Filtering can help showcase strong correlations between hidden data patterns. b. KZ filtering keeps the number of observations and the tail of a data set intact, preserving statistical validity.

RESULTS

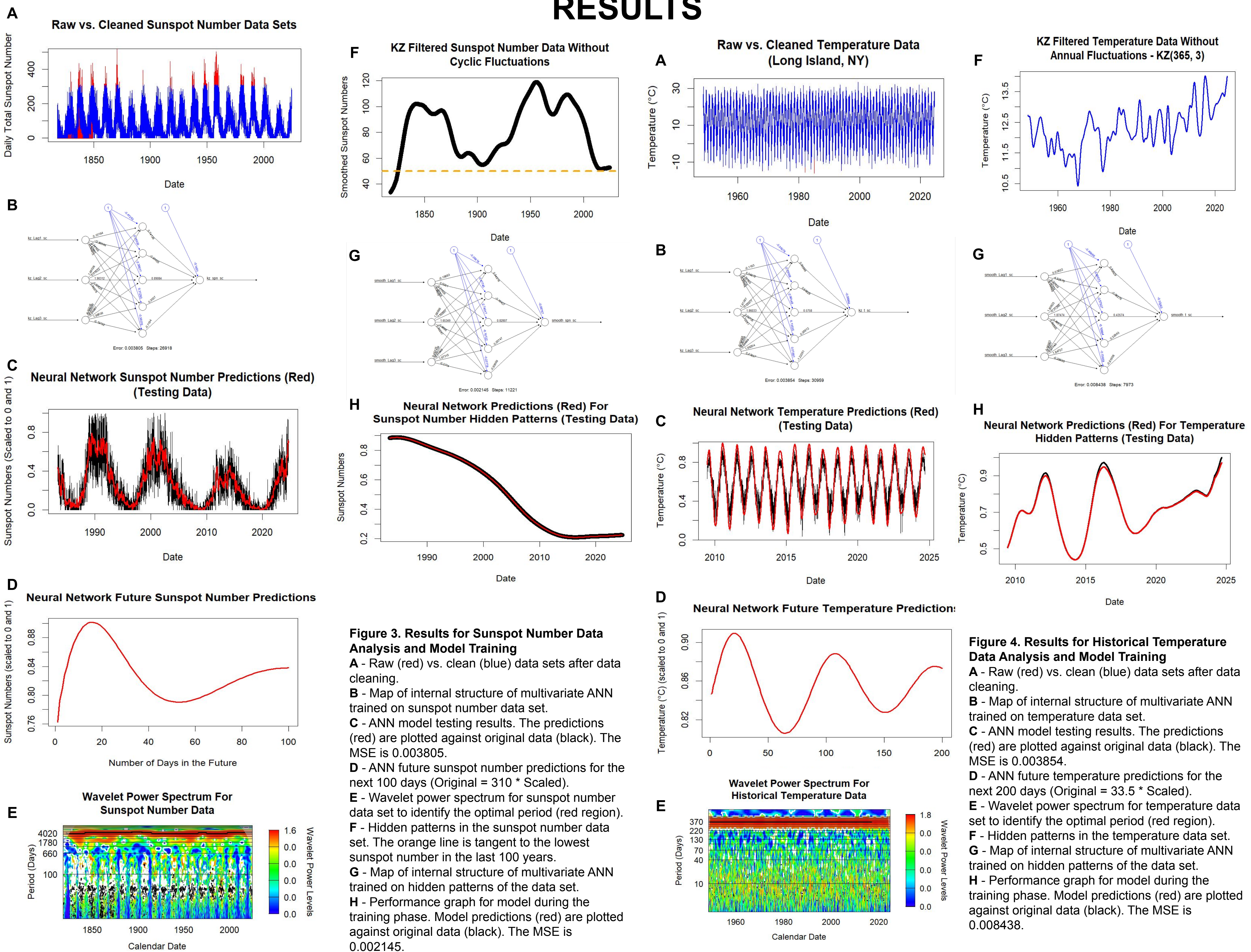


Figure 3. Results for Sunspot Number Data Analysis and Model Training
A - Raw (red) vs. clean (blue) data sets after data cleaning.
B - Map of internal structure of multivariate ANN trained on sunspot number data set.
C - ANN model testing results. The predictions (red) are plotted against original data (black). The MSE is 0.003805.
D - ANN future sunspot number predictions for the next 100 days (Original = 310 * Scaled).
E - Wavelet power spectrum for sunspot number data set to identify the optimal period (red region).
F - Hidden patterns in the sunspot number data set. The orange line is tangent to the lowest sunspot number in the last 100 years.
G - Map of internal structure of multivariate ANN trained on hidden patterns of the data set.
H - Performance graph for model during the training phase. Model predictions (red) are plotted against original data (black). The MSE is 0.002145.

Figure 4. Results for Historical Temperature Data Analysis and Model Training

A - Raw (red) vs. clean (blue) data sets after data cleaning.
B - Map of internal structure of multivariate ANN trained on temperature data set.
C - ANN model testing results. The predictions (red) are plotted against original data (black). The MSE is 0.003854.
D - ANN future temperature predictions for the next 200 days (Original = 33.5 * Scaled).
E - Wavelet power spectrum for temperature data set to identify the optimal period (red region).
F - Hidden patterns in the temperature data set.
G - Map of internal structure of multivariate ANN trained on hidden patterns of the data set.
H - Performance graph for model during the training phase. Model predictions (red) are plotted against original data (black). The MSE is 0.008438.